



NUMBER SYSTEM

Binary Value: (0,1)

1 octet = 8 bits / 1 byte = 8 bits

Decimal to Binary Conversion

74 01001010

21 00010101

82 01010010

248 11111000

100 01100100

255 11111111

150 10010110

0 00000000

128 10000000

192 11000000

224 11100000

240 11110000

241 11110001

120 01111000

250 11111010

254 11111110

0 0 0 128 64 32 16 8 4 2 1 — 1 octet = 8 bits

Hexa to Binary Conversion

1 Hexa = 4 bits

8 4 2 1

A 1010 0000 0000

FFF 1111 1111 1111

8AF 1000 1010 1111

FAB 1111 1010 1011

1C0 0001 1100 0000

CC0 1100 1100 0000

FF0 1111 1111 0000

F10 1111 0001 0000

AAA 1010 1010 1010

221 0010 0010 0001

ABC 1010 1011 1100

CB5 1100 1011 0101

42D 0100 0010 1101

48F 0100 1000 1111

2D0 0010 1101 0000

99D 1001 1001 1101

1, Types of mediums ?

three type of mediums

- * Radio Frequency — Wi-Fi (RF)
- * Electrical signal — Copper (UDP)
- * Light signal — Fiber optical (time)

2, Write down switch 8-pins ?

- 1 +T white orange
- 2 -T orange
- 3 +R white Green
- 4 +V Blue
- 5 -V white Blue
- 6 -R Green
- 7 x white Brown
- 8 x Brown

T-568A — T568B [Cross]

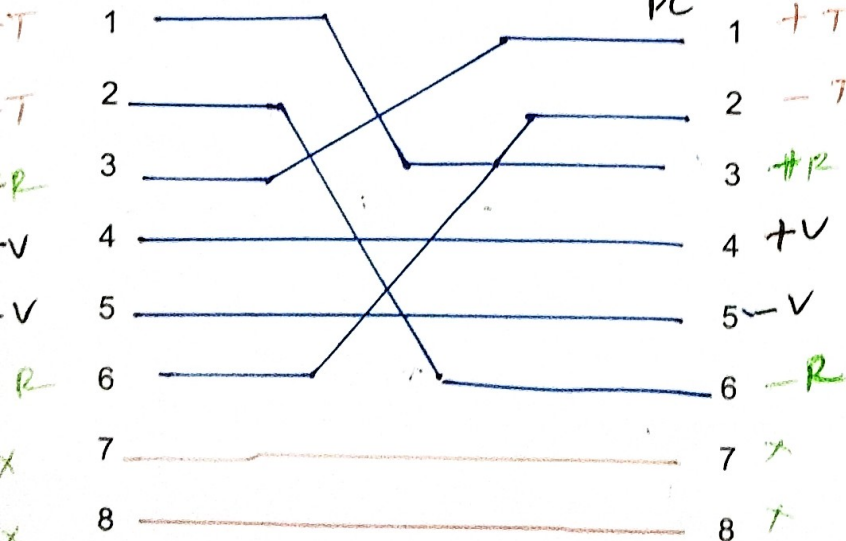
T-568B — T568B [Straight]

3, Connect utp cable between two devices

Router



PC



Device	+T	-T	+R	-R
Router	1	2	3	6
PC	1	2	3	6

It is two devices is Router & PC (UDP) cable crany header
It is used same function
(+T-T) (+R-R) It's connected
Crossover cable

T568A — T568B
(header) (header)

(without Auto MDI-X)

4, Explain AUTO MDIX ?

AUTO MDIX (Automatic medium Dependent Interface Crossover)
It is automated ethernet modern port automatically detected the type of devices connecting data follow assigned or changed straight (or) cross (or) cross to straight) transmit data.

5, Explain POE injector ?

POE injector provides the power of poe devices some devices need power but are installed where no power outlet is available. A poe injector supplies the power over ethernet cables.

6, Explain POE switch ?

POE switch (Power over ethernet) switch is overall consumption of power saved with poe devices. Data and power in single ethernet cables. no need additional (power and injector) It is provided single network connectivity and power of poe device like (wifi camera)

7, Explain types of nic & function ?

type of Nic: wired, wireless, network interface card = mac + ip

Functions:-

It is used to convert binary signals to electrical
Electrical signals to binary.